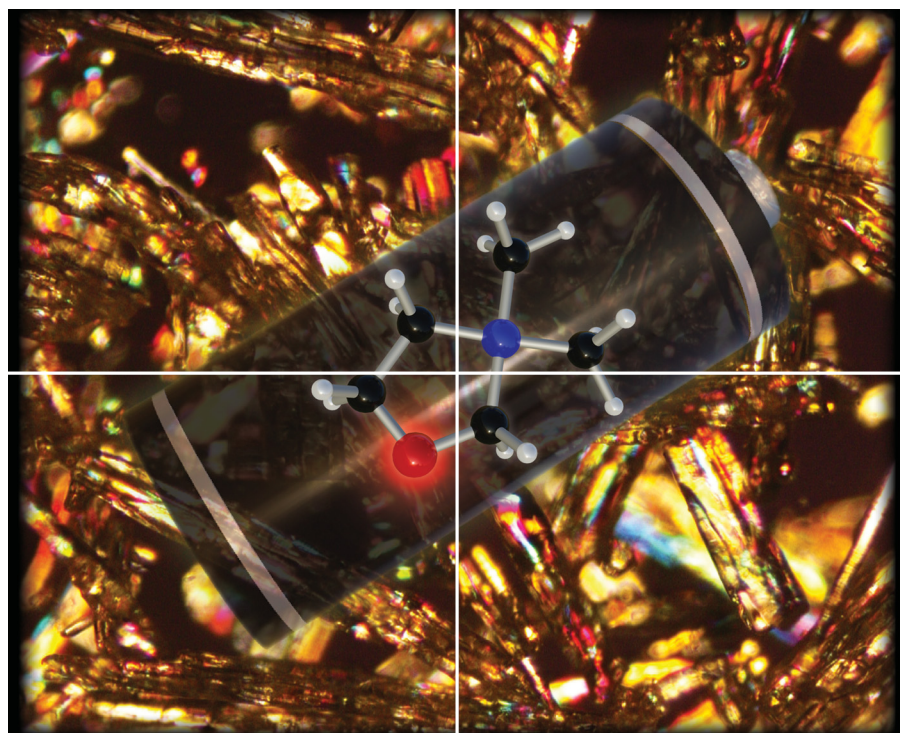




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Ionic liquids and plastic crystals utilising the oxazolidinium cation: the effect of ether functionality in the ring†

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The development of safer electrolytes is critical for the advancement of electrochemical energy storage applications. Ionic Liquids (ILs) and Organic Ionic Plastic Crystals (OIPCs) are attractive choices as electrolytes due to their high thermal stabilities and intrinsic ionic conductivities. However, fundamental understanding of the impact of cation and anion choice, to form ILs or OIPCs, is still lacking, particularly with regard to how this can affect the electrolyte properties. In this work, we explore the synthesis and characterisation of several oxazolidinium-based salts to understand the effect of the ether functional group in the cation ring. Specifically, we investigate the 3,3-dimethyloxazolidinium ([C₁moxa]⁺) cation, paired with dicyanamide ([DCA][−]), bis(fluorosulfonyl)imide ([FSI][−]), bis(trifluoromethanesulfonyl)imide ([TFSI][−]), tetrafluoroborate ([BF₄][−]) and hexafluorophosphate ([PF₆][−]) anions. In addition to investigating the influence of the anion, the effect of changing the cation side chain is examined with 3-ethyl-3-methyloxazolidinium bis(fluorosulfonyl)imide ([C₂moxa][FSI]). Studies of thermal behaviour revealed that these oxazolidinium-based salts can be highly disordered and exhibit wide phase I temperature ranges. The transport properties of the salts were determined by examining the ionic conductivity, ion dynamics using solid-state Nuclear Magnetic Resonance (NMR) analysis, and the free volume through Positron Annihilation Spectroscopy (PALS). It was found that oxazolidinium-based salts exhibit faster ion transport than the equivalent pyrrolidinium-based salts, which is consistent with the larger free volume. Hence, oxazolidinium-based salts demonstrate scope as promising electrolytes for battery applications due to their high ionic conductivity and good thermal and electrochemical stability.

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Introduction

The ever-increasing demand for portable high-energy devices necessitates a drive toward safer battery technologies. With the desire to move away from traditional flammable electrolytes, Ionic Liquids (ILs) and Organic Ionic Plastic Crystals (OIPCs) are promising candidates as safe and long-lasting electrolytes due to their high thermal stability and negligible vapour pressure. Both ILs and OIPCs, comprised entirely of ions, can also exhibit other beneficial properties such as wide electrochemical stability windows, intrinsic conductivity and chemical

stability.^{1,2} The distinction of ILs is that they are liquid at ambient temperatures and therefore offer higher ionic conductivities than neat OIPCs. On the other hand, OIPCs exist in the solid state, with the benefit of mitigating leakage problems, and combining OIPCs with lithium or sodium salts can lead to several orders of magnitude increase in ionic conductivity.³

OIPCs typically display one or more solid–solid transitions that lead to increased disorder of the solid prior to the melt, which as a result have a low entropy of fusion (ΔS_f). Such disorder is believed to give rise to inherent vacancies and defects within the OIPC structure, allowing for rotational/translational motions of the cations and anions.⁴ Moreover, OIPCs exhibit soft mechanical properties that enable them to be readily pliable without significant deformation or cracking.⁵ This intrinsic plasticity is advantageous for electrochemical device applications as it minimises issues of poor contact between the electrolyte and electrode in the case that volume expansion occurs.⁶ Indeed, as a result of the abovementioned properties, both ILs and OIPCs show considerable promise for use in lithium and sodium battery applications.¹

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The number of known ILs and OIPCs is steadily increasing, with the selection of anions including imides,⁷ borates,⁸ phosphates,⁹ sulfonates,¹⁰ fluorides,¹¹ and thiocyanates.¹² Whilst the choice of anion can certainly lead to differences in the thermal and physical properties of these salts, the trends are not necessarily straightforward or predictable. For example, Moriya *et al.* observed that the restricted flexibility of the cyclic perfluorosulfonylamide ([CPFSA][−]) anion, compared to the bis(trifluoromethanesulfonyl)imide ([TFSI][−]) anion (aka [NTf₂][−]), can lead to an increase in melting point when paired with the *N*-ethyl-*N*-methylpyrrolidinium ([C₂mpyr]⁺) cation (302 vs. 86 °C respectively).¹³ On the other hand, Yamaguchi *et al.* found that [C₂mpyr][BETI] (BETI = bis[(pentafluoroethyl) sulfonyl]imide) melts at 88 °C, similar to [C₂mpyr][TFSI], despite its additional modes of rotation (*i.e.* longer flexible fluoroalkyl chain).⁷

Furthermore, many organic cation families have been investigated to understand their influence on the physicochemical properties of the ILs/OIPCs, including pyrrolidinium,^{14,15} phosphonium,¹⁶ ammonium,^{17,18} sulfonium¹⁹ and guanidinium²⁰ moieties. Studies into structural variants of these cations can provide some insight into the nature of these materials. For example, a branched isopropyl substituent on the *N*-isopropyl-*N*-methylpyrrolidinium ([C₁₃mpyr]⁺) cation tends to favour solid formation (*i.e.* higher lattice energy) compared to the diethyl substituents on the symmetrical *N,N*-diethylpyrrolidinium ([C₂epyr]⁺) cation; despite being structural isomers, [C₁₃mpyr][FSI] and [C₂epyr][FSI] melt at notably different temperatures of 189 °C and 131 °C respectively.^{15,21} Furthermore, Yamada *et al.* found that longer alkyl side chains on the pyrrolidinium cation of an OIPC led to a decrease in the melting point ([C₂epyr][FSI] < [C₂mpyr][FSI] < [C₁mpyr][FSI]).²² Moreover, [C₂epyr][FSI] exhibits higher solid-state ionic conductivity, indicating more disorder with this particular OIPC.¹⁵

Many studies have demonstrated an improvement in the ionic conductivity of ILs by introducing ether functionality onto the alkyl side chain of the cation.^{23–26} Furthermore, incorporation of these aliphatic ether oxygens on the cation has been shown to decrease the lithium solvation number (*i.e.* reducing the overall number of interactions between Li⁺ ions and [TFSI][−] anions).^{27,28} This, in addition to the reduced viscosity, was shown to increase the mobility and transference number of Li⁺ ions, and subsequently resulted in better rate capability performance.^{29,30} Hence, there are some proven benefits of utilising ether functional groups in the side chain of cation. Less explored, however, is the incorporation of an ether group into the ring of the cation (*i.e.* cyclic ether) such as the family of oxazolinium, oxazolidinium and morpholinium cations, particularly for the development of new OIPCs. Although oxazolinium cations tend to undergo ring opening under acidic conditions,³¹ both oxazolidinium and morpholinium cations have been shown to be stable and have been synthesised under acid–base quarterisation conditions by Zhou *et al.*³² Although morpholinium-based ILs have been studied,^{33,34} oxazolidinium-based salts exhibit a relatively higher degree of ionic conductivity and thus are of current interest.³²

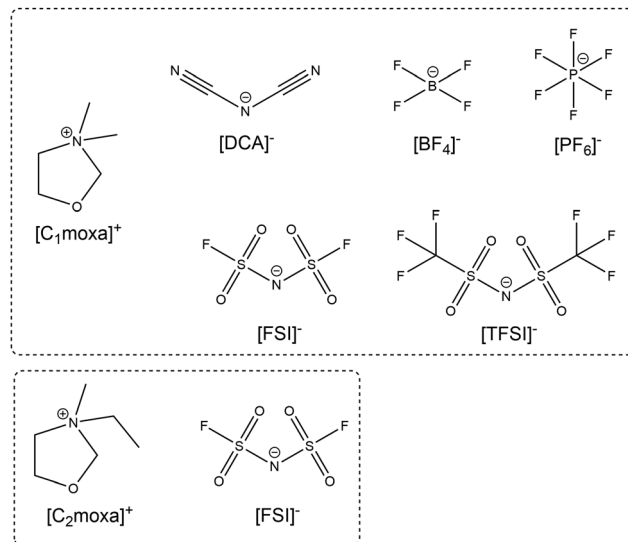


Fig. 1 Molecular structure of [C₁moxa]⁺ and [C₂moxa]⁺ cations paired with various anions investigated in this study.

The oxazolidinium cation is structurally similar to the pyrrolidinium cation, but with an ether functional group replacing the beta –CH₂– group in the 5-membered ring (see Fig. 1). In the work carried out by Zhou and co-workers, they observed a small decrease in the ionic conductivity of the IL [C₄moxa][TFSI] compared to [C₄mpyr][TFSI].³² Similar trends in the ionic conductivity were found when comparing [C₄moxa]⁺-based ILs paired with perfluoroborate anions such as [BF₄][−], [CF₃BF₃][−] and [C₂F₅BF₃][−]. On the other hand, the lower Δ*S*_f with [C₄moxa][BF₄] compared to [C₄mpyr][BF₄] (12 vs. 24 J K mol^{−1} respectively), suggests that solid oxazolidinium-based salts exhibit a higher degree of disorder than the pyrrolidinium analogues. However, this approach has not been widely explored for the development of new OIPCs, reflecting the fact that the range of known OIPC-forming cations is currently much smaller than for ILs. Furthermore, the potential impact of this cyclic ether group on OIPCs in terms of the thermal phase behaviour, transport properties, conduction mechanism, and its coordination with Li salt, is not yet understood.

Here we report the synthesis and detailed characterisation of a range of oxazolidinium-based OIPCs and examine the impact of cation and anion structure on the thermal behaviour, transport properties and electrochemical stability. [C₁moxa][DCA] was found to be a room temperature IL, whereas [C₁moxa][BF₄], [C₁moxa][PF₆], [C₁moxa][FSI] and [C₂moxa][FSI] are OIPCs with wide plastic crystal temperature ranges. The effect of the alkyl chain length on the oxazolidinium cation was also explored through the comparative analysis of [C₁moxa][FSI] and [C₂moxa][FSI]. The new oxazolidinium-based salts show higher solid-state ionic conductivities, narrower NMR linewidths (*i.e.* more mobile ionic species) and larger free volumes than the equivalent pyrrolidinium-based salts, reflecting the beneficial impact of the ether group in the ring.

Experimental

Materials and methods

3,3-Dimethyloxazolidinium iodide (97%, Boron Molecular, Australia), 3-methyloxazolidine ($\geq 97\%$, Boron Molecular), methyl 4-methylbenzenesulfonate ($\geq 98\%$, Boron Molecular), ethyl bromide (98%, Sigma Aldrich), ethyl acetate (HPLC grade, Chem-Supply, Australia), isopropanol (99.5%, Acros Organics), anhydrous acetonitrile (CH_3CN) ($\geq 99.9\%$, Sigma Aldrich), lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) (99.9%, Solvay, Canada), potassium bis(fluorosulfonyl)imide (KFSI) (99.9%, Suzhou Fluolyte Co., China), sodium iodide (NaI) (99%, Sigma Aldrich, Australia), dichloromethane (HPLC grade, lab supply), silver tetrafluoroborate (AgBF_4) ($\geq 99.99\%$, Oakwood Chemical, USA), silver hexafluorophosphate (AgPF_6) (98%, Oakwood Chemical, USA), sodium dicyanamide (NaDCA) (96%, Sigma Aldrich, Australia) and silver nitrate (AgNO_3) (LR, Chem-Supply, Australia) were used as received. The molecular structures and abbreviations are shown in Fig. 1. Unless specified, synthesis was carried out under Argon (Ar) atmosphere. All samples were dried under high vacuum for at least 48 hours prior to characterisation.

Solution-state ^1H , ^{13}C and ^{19}F Nuclear Magnetic Resonance (NMR) spectra were measured on a Bruker Avance III instrument operating at 400 MHz, 100 MHz and 375 MHz respectively, using deuterated methanol (CD_3OD), dimethylsulfoxide ($(\text{CD}_3)_2\text{SO}$), acetone ($(\text{CD}_3)_2\text{CO}$) or deuterated water (D_2O). Mass spectrometry was performed on an Agilent 1200 series HPLC system. Samples were sent to the Campbell Microanalytical Laboratory, New Zealand and Monash Analytical Platform for elemental analysis. The residual silver and iodide contents were determined by an Ionode IJ-Ag, IJ-I, or IJ-Br ion selective electrode (ISE), after calibration with 10 and 100 ppm solutions of the silver (from silver nitrate), iodide ion (sodium iodide) or bromide ion (sodium bromide). Quantification of lithium and potassium was carried out using an inductively coupled plasma-mass spectrometer (ICP-MS; NexION 350X, PerkinElmer, USA). Internal standards Sc (200 ppb) and Rh (20 ppb) in 1% aqua regia were used for correction of matrix effects. The internal standard solution was mixed prior to the nebulizer using a T-piece in a 1:1 ratio. Calibration standards for lithium and potassium (PerkinElmer, silver, lithium or potassium standard 1000 ppm in 2% HNO_3) were prepared at 0.1, 1, 10, 50, 100 and 500 ppb with 2% suprapur HNO_3 in each. The mass spectrometer was operated in kinetic energy discrimination mode (KED) with 50 ms dwell times, 20 sweeps, one reading and three replicates. The plasma source conditions were: nebulizer gas flow 1.02 L min^{-1} , auxiliary gas flow 1.2 L min^{-1} , plasma gas flow 15 L min^{-1} , and ICP RF power 1500 W. Data analysis was carried out using Syngistix (PerkinElmer) software. Signal responses were normalised to the scandium internal standard.

Manual melting points were determined visually using a Gallenkamp melting point apparatus prior to Differential Scanning Calorimetry (DSC) measurements. DSC was carried out on a Mettler Toledo DSC STARE at a scan rate of 10 $^\circ\text{C min}^{-1}$ for heating and cooling cycles. Three heating and cooling cycles

were performed under a N_2 atmosphere. Thermogravimetric analysis (TGA) was performed on a Mettler Toledo TGA/DSC 1 STARE System over a range of 25 to 550 $^\circ\text{C}$ under N_2 (30 mL min^{-1}), at a heating rate of 10 $^\circ\text{C min}^{-1}$. The samples were prepared in aluminium pans.

The ionic conductivity was measured using electrochemical impedance spectroscopy with a Biologic MTZ35 Solartron Modulab. The conductivity of liquids or soft solids were measured by using a dip cell *via* two platinum (Pt) electrodes, with a temperature range from 30 to 90 $^\circ\text{C}$, using 1 mM KCl solution as a reference standard at 30 $^\circ\text{C}$. The conductivities of solid samples, pressed into a pellet, were measured from 30 to 90 $^\circ\text{C}$ using a barrel cell.

Solid-state NMR was measured on a Bruker Avance III widebore NMR spectrometer, operating at 500 and 470 MHz for ^1H and ^{19}F respectively, equipped with a 4 mm HF/X double resonance probe. The samples were packed into 4 mm ZrO_2 NMR rotors in an Ar-filled glovebox, sealed with a ZrO_2 ceramic cap before being transferred out and inserted into the probe for variable temperature measurements. ^1H static spectra were recorded with a single pulse excitation, whereas the ^{19}F static spectra were recorded with Hahn echo using echo delay of 10 μs . Recycle delay was 10 s for both ^1H and ^{19}F experiments.

The average pore size and relative quantity was investigated using Positron Annihilation Lifetime Spectroscopy (PALS). The samples were measured using EG&G Ortec spectrometers set in coincidence with a resolution of 230 ps. Two 1 mm thick disk samples were placed on both sides of the positron point source (1.8 MBq $^{22}\text{NaCl}$ sealed in a Mylar envelope) and measured for a minimum of five files each with 4.5×10^6 integrated counts. The spectra were analysed using LTV9 software³⁵ and were fitted to three lifetime components. The first lifetime was fixed to 125 ps and attributed to the annihilation of *para*-positronium (pPs; bound state of the positron and an electron of opposite spin). The second lifetime was approximated to 400 ps and attributed to free annihilation of the positron with free electrons within the samples. The final lifetime component is the long lifetime component related to the annihilation of the bound state of the positron with an electron of the same spin, *ortho*-positronium, oPs. The oPs lifetime, τ_3 was used to calculate the average free volume radius (r) using the Tao-Eldrup equation:^{36,37}

$$\tau_3 = \frac{1}{2} \left(1 - \frac{r}{r + \Delta r} + \frac{1}{2\pi} \sin \left[\frac{2\pi r}{r + \Delta r} \right] \right)^{-1}$$

here, Δr is approximated to 1.66 Å as the empirical measurement for the electron layer thickness.

Cyclic Voltammetry (CV) experiments were conducted using a three-electrode cell, a Pt disk working electrode, a coiled Pt wire counter electrode, and a Pt wire pseudo-reference electrode. Experiments were carried out in an Argon-filled glovebox, using a Biologic SP-200 potentiostat.

Synthetic procedures

3,3-Dimethyloxazolidinium 4-methylbenzenesulfonate, [C_1moxa] [Ts]. To a solution of 3-methyloxazolidine (10.00 g, 115 mmol) in

acetone at 50 °C, a solution of methyl 4-methylbenzenesulfonate (21.38 g, 115 mmol) in acetone was added dropwise. After the exothermic reaction settled, the mixture was heated to reflux with stirring overnight under nitrogen (N₂) atmosphere. After cooling to –30 °C with stirring, the solid which separated was filtered off, rinsed with cold acetone, and dried at 70 °C *in vacuo* (ca. 0.1 mbar) to afford an off-white solid (24.60 g, 61%). ¹H NMR (200 MHz, (CD₃)₂SO): 2.29 ppm (s, –ArCH₃, 3H), 3.17 ppm (s, N(CH₃)₂, 6H), 3.68 ppm (t, *J*_{HH} = 7.4 Hz, NCH₂CH₂O, 2H), 4.24 ppm (t, *J*_{HH} = 7.2 Hz, NCH₂CH₂O, 2H), 4.82 ppm (s, NCH₂O, 2H), 7.12 ppm (d, *J*_{HH} = 8.0 Hz, ArCH, 2H), 7.49 ppm (d, *J*_{HH} = 8.0 Hz, ArCH, 2H).

3-Ethyl-3-methyloxazolidinium bromide, [C₂moxa][Br]. To a solution of 3-methyloxazolidine (100.00 g, 1.15 mol) in ethyl acetate (500 mL) at 40 °C, ethyl bromide (143.84 g, 1.32 mol) was added dropwise over one hour. The mixture was stirred under reflux overnight. Ethyl acetate was removed *in vacuo* (ca. 0.1 mbar), at 60 °C, to afford a light brown oil (191.00 g, 85%). ¹H NMR (200 MHz, (D₂O): 1.47 ppm (t, *J*_{HH} = 7.4 Hz –NCH₂CH₃, 3H), 3.22 ppm (s, NCH₃, 3H), 3.68 ppm (q, *J*_{HH} = 7.4 Hz, NCH₂CH₃, 2H), 3.84 ppm (t, *J*_{HH} = 7.4 Hz, NCH₂CH₂O, 2H), 4.40 ppm (t, *J*_{HH} = 7.4 Hz, NCH₂CH₂O, 2H), 4.92 ppm (q, NCH₂O, 2H).

3,3-Dimethyloxazolidinium dicyanamide, [C₁moxa][DCA]. Silver dicyanamide (AgDCA) (21.84 g, 126 mmol) was prepared from a literature method³⁸ and immediately added to water (100 mL) and [C₁moxa][I] (9.30 g, 42 mmol), followed by stirring overnight, in the dark. The reaction mixture was filtered under vacuum followed by further purification through Celite to remove silver iodide particles. The resulting filtrate was dried *in vacuo* (ca. 0.1 mbar) at 55 °C for one week to afford a colourless liquid [C₁moxa][DCA] (6.50 g, 93%). ¹H NMR (400 MHz, CD₃OD): 3.29 (s, N(CH₃)₂, 6H), 3.78 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.39 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.89 (s, NCH₂O, 2H); ¹³C NMR (100 MHz, CD₃OD): 49.3 (N(CH₃)₂), 61.6 (NCH₂CH₂), 66.3 (NCH₂CH₂), 93.6 (NCH₂O), 119.2 (CN) ppm. ES⁺ *m/z* 101.5 (C₅H₁₂NO)⁺, ES[–] *m/z* 65.9 (C₂N₃)[–]. Iodide content (ISE) <10 ppm. Silver content (ISE) <10 ppm. Water content was 640 ppm.

3,3-Dimethyloxazolidinium bis(trifluoromethanesulfonyl) imide, [C₁moxa][TFSI]. [C₁moxa][I] (6.14 g, 27 mmol) was dissolved in water (10 mL) and added to a solution of LiTFSI (7.73 g, 27 mmol) and water (20 mL). Upon mixing the above solutions, a white precipitate formed instantaneously. The mixture was heated to dissolve the insoluble salt and was left in a fridge overnight to allow for recrystallisation. These crystals were filtered and further re-crystallised twice. The crystals were dried *in vacuo* (ca. 0.1 mbar) for 24 hours at 50 °C to afford a white solid (6.00 g, 59%). ¹H NMR (400 MHz, CD₃OD): 3.27 (s, N(CH₃)₂, 6H), 3.75 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.38 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.85 (s, NCH₂O, 2H); ¹³C NMR (100 MHz, CD₃OD): 48.2 (N(CH₃)₂), 61.5 (NCH₂CH₂), 66.2 (NCH₂CH₂), 93.6 (NCH₂O), 119 (CF₃, *J*_{CF} = 320 Hz) ppm. ¹⁹F NMR (375 MHz, CD₃OD): –80.66 (CF₃) ppm. ES⁺ *m/z* 101.5 (C₅H₁₂NO)⁺, ES[–] *m/z* 279.9 (NS₂O₂C₂F₆)[–]. Anal. calculated for C₇H₁₂N₂F₆O₅S₂; C, 21.99; H, 3.16; N, 7.33. Found: C, 21.28; H, 3.10; N, 7.11. Iodide

content (ISE) <10 ppm. Lithium (ICP-MS) 105 ppm. The melting point measured manually was 74 °C.

3,3-Dimethyloxazolidinium bis(fluoromethanesulfonyl)imide, [C₁moxa][FSI]. Potassium bis(fluorosulfonyl)imide (19.25 g, 88 mmol) was added to a solution of 3,3-dimethyloxazolidin-3-ium 4-methylbenzenesulfonate (24.00 g, 84 mmol) and water (70 mL), and left to stir overnight under N₂. Once the water was removed *in vacuo*, the crude solid was dissolved in methanol. After cooling in an ice-bath, a white solid was filtered off and rinsed with cold methanol. To the filtrate, water was added followed by cooling to –35 °C and left to stir. The precipitate was filtered and rinsed with cold methanol (–35 °C) followed by drying *in vacuo* (ca. 0.1 mbar) overnight to afford an off-white solid (9.00 g, 36%). ¹H NMR (200 MHz, (CD₃)₂SO): 3.16 ppm (s, N(CH₃)₂, 6H), 3.67 ppm (t, *J*_{HH} = 7.2 Hz, NCH₂CH₂O, 2H), 4.25 ppm (t, *J*_{HH} = 7.4 Hz, NCH₂CH₂O, 2H), 4.79 ppm (s, NCH₂O, 2H). ¹⁹F NMR (375 MHz, (CD₃)₂CO): 51.6 (F) ppm. ES⁺ *m/z* 102.0 (C₅H₁₂NO)⁺ ES[–] *m/z* 179.9 (NS₂O₂F₂)[–]. Anal. calculated for C₅H₁₂N₂F₂O₅S₂; C, 21.27; H, 4.29; N, 9.92. Found: C, 21.25; H, 4.37; N, 9.87. Potassium content (ICP) <10 ppm. The melting point measured manually was 207 °C.

3,3-Dimethyloxazolidinium hexafluorophosphate, [C₁moxa][PF₆]. [C₁moxa][I] (4.65 g, 20 mmol) was dissolved in anhydrous acetonitrile and added to a solution of AgPF₆ (5.13 g, 20 mmol) and acetonitrile. Upon mixing the colourless solutions, a yellow precipitate of silver iodide formed instantaneously. The solution was stirred under Argon (Ar) atmosphere for one hour in the dark at 50 °C. The warm solution was filtered through a syringe filter (0.22 µm, PTFE). After acetonitrile was removed *in vacuo*, the white solid was dissolved in acetone (20 mL) and placed in the fridge overnight to precipitate black particles out. The solution was filtered through Celite and dried *in vacuo* (ca. 0.1 mbar) to afford a white solid (3.50 g, 70%). ¹H NMR (400 MHz, (CD₃)₂SO): 3.17 (s, N(CH₃)₂, 6H), 3.67 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.25 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.80 (s, NCH₂O, 2H); ¹³C NMR (100 MHz, (CD₃)₂SO): 49.8 (N(CH₃)₂), 61.4 (NCH₂CH₂), 66.5 (NCH₂CH₂), 93.2 (NCH₂O) ppm. ¹⁹F NMR (375 MHz, (CD₃)₂SO): –71.1, –68.1 (F) ppm. ES⁺ *m/z* 102.0 (C₅H₁₂NO)⁺, ES[–] *m/z* 144.9 (PF₆)[–]. Anal. calculated for C₅H₁₂NOF₆P; C, 24.30; H, 4.89; N, 5.66. Found: C, 23.66; H, 4.81; N, 5.30. Iodide content (ISE) <10 ppm. Silver content (ISE) 150 ppm.

3,3-Dimethyloxazolidinium tetrafluoroborate, [C₁moxa][BF₄]. [C₁moxa][I] (3.82 g, 16 mmol) was dissolved in anhydrous acetonitrile and added to a solution of AgBF₄ (3.25 g, 16 mmol) and acetonitrile. Upon mixing, a yellow precipitate of silver iodide formed instantly where the solution was left to stir under Ar for two hours in the dark at 50 °C. The warm solution was filtered through a syringe filter (0.22 µm, PTFE) and acetonitrile was removed *in vacuo* (ca. 0.1 mbar) to afford a white solid (2.7 g, 90%). ¹H NMR (400 MHz, CD₃OD): 3.27 (s, N(CH₃)₂, 6H), 3.75 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.37 (t, *J*_{HH} = 7 Hz, NCH₂CH₂, 2H), 4.86 (s, NCH₂O, 2H); ¹³C NMR (100 MHz, CD₃OD): 49.2 (N(CH₃)₂), 61.5 (NCH₂CH₂), 66.3 (NCH₂CH₂), 93.6 (NCH₂O) ppm. ¹⁹F NMR (375 MHz, CD₃OD): –153.9, –153.7 (F) ppm. ES⁺ *m/z* 102.0 (C₅H₁₂NO)⁺, ES[–] *m/z* 86.8 (BF₄)[–]. Anal. calculated

for $C_5H_{12}NF_4B$; C, 31.78; H, 6.40; N, 7.41. Found: C, 32.16; H, 5.93; N, 7.24. Iodide content (ISE) 330 ppm. Silver content (ICP-MS) 50 ppm.

3-Ethyl-3-methyloxazolidinium bis(fluoromethanesulfonyl) imide, $[C_2moxa][FSI]$. To a solution of 3-ethyl-3-methyloxazolidinium bromide (70.00 g, 357 mmol) in water (200 mL), a solution of potassium bis(fluorosulfonyl)imide (78.05 g, 356 mmol) in water (150 mL) was added and stirred overnight at room temperature. Once the water was removed *in vacuo*, at 70 °C, ethanol was added and left to stir for one hour; the solid which separated was filtered off and washed with ethanol. The filtrate was concentrated *in vacuo*. The residue was then treated with cold water and left to stir overnight. The solid which separated was filtered off, washed with cold water and left to dry *in vacuo*. Finally, pentane was added and left to stir, followed by filtration and drying *in vacuo* (*ca.* 0.1 mbar) to afford an off-white solid (85.00 g, 80%). 1H NMR (200 MHz, $(CD_3)_2SO$): 1.30 ppm (t, $J_{HH} = 7.2$ Hz, NCH_2CH_3 , 3H), 3.07 ppm (s, NCH_3 , 3H), 3.45 ppm (q, $J_{HH} = 7.2$ Hz, NCH_2CH_3 , 2H), 3.65 ppm (m, NCH_2CH_2O , 2H), 4.22 ppm (m, NCH_2CH_2O , 2H), 4.84 ppm (q, $J_{HH} = 7.6$ Hz, NCH_2O , 2H). ^{19}F NMR (375 MHz, $(CD_3)_2CO$): 51.5 (F) ppm. $ES^+ m/z$ 116.0 ($C_6H_{14}NO$) $ES^- m/z$ 179.9 ($NS_2O_2F_2$). Anal. calculated for $C_5H_{12}N_2F_2O_5S_2$; C, 24.40; H, 4.44; N, 9.49. Found: C, 24.25; H, 5.00; N, 9.41. Bromide content (ISE) 450 ppm. Potassium content (ICP) 500 ppm. The manual melting point obtained was 130–132 °C.

Results and discussion

Thermal properties

The Differential Scanning Calorimetry (DSC) traces of the oxazolidinium-based salts are shown in Fig. 2, with the phase transition temperatures and entropy changes presented in Table 1. For all the oxazolidinium-based salts investigated, at least one large entropy change solid–solid phase transition is evident prior to melting. These phases are denoted as phase I, II, III, *etc.* (from highest to lowest temperature phase) and they indicate significantly increased disorder of these salts with heating. While $[C_1moxa][DCA]$ displays one solid–solid

transition, it melts at 15 °C and exists as an IL at room temperature. The OIPCs $[C_1moxa][BF_4]$, $[C_1moxa][FSI]$, and $[C_2moxa][FSI]$ have entropy of fusion (ΔS_f) values below $20 J K^{-1} mol^{-1}$, and thus obey Timmerman's criteria for plastic crystal behaviour in molecular plastic crystals.³⁹ These OIPCs also display wide phase I ranges (> 160 °C) that, importantly, in most cases span typical device application temperatures. This is advantageous for their use as electrolytes as phase I is the most disordered and conductive phase. On the other hand, $[C_1moxa][TFSI]$ has a narrower phase I temperature range and exhibits a ΔS_f value above $20 J K^{-1} mol^{-1}$. It exhibits a relatively low melting point (for an OIPC) at 72 °C, and thus the addition of lithium salt (in order to obtain a lithium conducting electrolyte) may form quasi-solid state or low melting eutectic mixtures, depending on the concentration, for solid or liquid-state battery applications.^{21,40} Whilst $[C_1moxa][PF_6]$ does not show rich phase behaviour, due to decomposition before the melting phase, $[C_1moxa][BF_4]$ undergoes three phase transitions after a small transition near -50 °C. The relative enthalpies of the low temperature transitions for $[C_1moxa][BF_4]$ depend on the thermal history of the sample, as is often observed for OIPCs: the first DSC cycle shows a larger transition at -50 °C (see Fig. S1a, ESI†) but then a lower enthalpy for the transition at -7 °C than is seen in the second DSC cycle shown here. However, the ΔS_f is consistent across three DSC cycles.

In addition to the range of $[C_1moxa]^+$ -based salts discussed above, $[C_2moxa][FSI]$ was explored to examine the effect of an ethyl substituent on the oxazolidinium cation and to initially assess the feasibility of $[C_2moxa]^+$ -based salts. The choice of the $[FSI]^-$ anion was driven by the tendency of such OIPCs to form highly plastic materials and exhibit low ΔS_f values.^{18,41} As shown in Fig. 2, $[C_2moxa][FSI]$ displays additional solid–solid phase transitions compared to $[C_1moxa][FSI]$ (and other $[C_1moxa]^+$ -based salts), which may be due to additional motions of the ethyl side chain. The DSC trace of $[C_2moxa][FSI]$ recorded using a heating rate of 2 °C min^{-1} (instead of 10 °C min^{-1}) shows slightly different phase behaviour and ΔS values, again reflecting a dependence on thermal treatment of these OIPCs (see Fig. S1b, ESI†). The comparison of ΔS_f values for $[C_1moxa][FSI]$ vs. $[C_2moxa][FSI]$ shows only a slight increase (4 vs. $7 J K^{-1} mol^{-1}$ respectively).

A similarly small change in ΔS_f has been reported upon increasing the alkyl chain from methyl to ethyl in $[C_1mpyr][FSI]$ vs. $[C_2mpyr][FSI]$ (14 vs. $11 J K^{-1} mol^{-1}$ respectively).²² This indicates only a small change in the disorder of the oxazolidinium OIPCs upon the change to an ethyl side chain, whereas the impact on melting point is much greater, as discussed below. As the initial analysis of $[C_2moxa][FSI]$ reveals a wide phase I range and high disorder (*i.e.* low ΔS_f), other $[C_2moxa]^+$ -based salts may also show considerable promise as OIPCs with attractive properties, and this will be further investigated in subsequent studies.

Interestingly, the oxazolidinium-based salts studied show considerable decreases in the melting point, and thus have lower lattice energy, compared to the equivalent pyrrolidinium-based salts (see Table S1, ESI†). Therefore, this change in cation

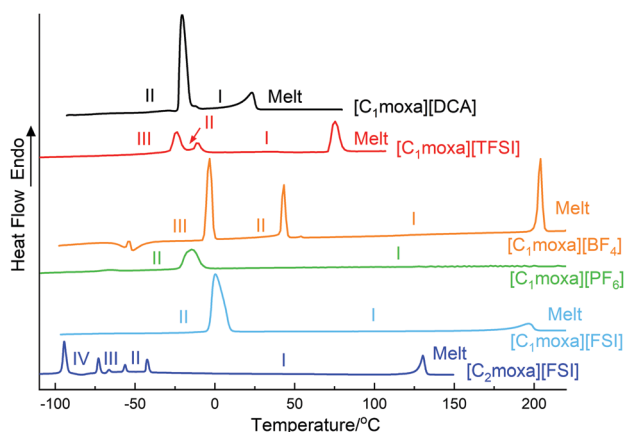


Fig. 2 DSC traces of $[C_1moxa]^+$ -based salts and $[C_2moxa][FSI]$.

Table 1 Phase transition temperatures, associated entropy changes and decomposition temperatures for [C₁moxa]⁺-based salts and [C₂moxa][FSI]

	V–IV		IV–III		III–II		II–I		I–melt		<i>T_d(onset)</i> /°C
	<i>T_{s-s}</i> /°C ±1 °C	ΔS_{s-s} /J K ⁻¹ mol ⁻¹ ±10%	<i>T_{s-s}</i> /°C ±1 °C	ΔS_{s-s} /J K ⁻¹ mol ⁻¹ ±10%	<i>T_{s-s}</i> /°C ±1 °C	ΔS_{s-s} /J K ⁻¹ mol ⁻¹ ±10%	<i>T_{s-s}</i> /°C ±1 °C	ΔS_{s-s} /J K ⁻¹ mol ⁻¹ ±10%	<i>T_m</i> /°C ±1 °C	ΔS_f /J K ⁻¹ mol ⁻¹ ±10%	
[C ₁ moxa][DCA]							–23	37	15	11	230
[C ₁ moxa][TFSI]					–28	20	–13	6	72	24	340
[C ₁ moxa][BF ₄]			–56	n/a	–7	27	41	11	202	12	323
[C ₁ moxa][PF ₆]							–22	25	>343	n/a	343
[C ₁ moxa][FSI]							–3	60	187	4	290
[C ₂ moxa][FSI]	–96	16	–74	5	–58	2	–44	4	127	7	280

can lead to distinct changes in phase behaviour *e.g.* at 30 °C, [C₁moxa][BF₄] and [C₁mpyr][BF₄] exist in phase II and III respectively, with melting points at 204 °C and >340 °C respectively.⁴² These oxazolidinium-based salts also exhibit lower ΔS_f values than equivalent pyrrolidinium-based salts (Table S1, ESI[†]), suggesting that they exhibit a greater degree of disorder prior to melting. Such differences may derive from a number of factors such as increased cation asymmetry and higher polarity due to the ether group on the ring. Furthermore, Henderson *et al.* observed that in imidazolium salts, adjacent ether atoms packed in such a way as to lie as far away from each other as possible; this resulted in significantly different ion crystal packing in 1-(2-methoxyethyl)-2,3-dimethylimidazolium [Cl][–] and [PF₆][–] salts compared to the analogous 1-butyl-2,3-dimethylimidazolium salts.⁴³ It is possible that similar differences in structure are present with these oxazolidinium-based OIPCs, and are reflected in the changes in thermal behaviour when an ether functional group is present in the ring.

Thermal gravimetric analysis

The thermal degradation temperature of salts is often used to estimate of the maximum operating temperature limit for use in practical applications. Therefore, an initial assessment of thermal stability through thermal gravimetric analysis was carried out to determine the decomposition temperature of [C₁moxa]⁺-based salts with various anions. In this work, the thermal stability is defined by the onset of decomposition *via* dynamic analysis under N₂ at a heating rate of 10 °C min^{–1} (*i.e.* the step-tangent method). The decomposition temperature values for the oxazolidinium-based salts are shown in Table 1, and the traces in Fig. S2 (ESI[†]).

Many of the oxazolidinium-based salts studied show good thermal stability. However, [C₁moxa][DCA] exhibits decomposition at relatively lower temperatures of 230 °C, as similarly observed with other [DCA][–]-based salts.⁴⁴ [C₁moxa][TFSI] displays higher thermal stability than [C₁moxa][FSI] (340 °C and 290 °C respectively), consistent with other studies where such anions were paired with other cation families such as imidazolium,⁴⁵ pyrrolidinium,²¹ phosphonium,¹⁶ and sulfonium.⁴⁶ This trend has previously been attributed to the higher nucleophilicity of the [FSI][–] anion, as [TFSI][–] exhibits increased electron delocalisation through its CF₃ substituents.⁴⁷ The [C₁moxa][PF₆] salt exhibits slightly higher thermal stability than the [C₁moxa][BF₄]. In contrast, [BF₄][–]-based salts are more thermally stable than [PF₆][–]-based salts when paired with cations such as

phosphonium,¹⁶ hexamethylguanidinium,²⁰ and imidazolium.⁴⁸ Therefore, the thermal stability of salts cannot be based on the trends of the anion alone, and one must also consider the effect of the cation.

Comparison of the thermal stability of the oxazolidinium-based salts to the equivalent pyrrolidinium-based salts reveals a small decrease in decomposition temperatures, of approximately 10–50 °C (see Table S1, ESI[†]). Detailed quantitative comparison is not feasible due to different heating scan rates and measurement conditions utilised by the other research groups. Nonetheless, the decrease in thermal stability is consistent with the report by Zhou *et al.*³² on oxazolidinium-based ILs, and with other studies where an ether group is incorporated into the alkyl side chain of IL cations (*vs.* a saturated alkyl chain).^{49,50} Nevertheless, these oxazolidinium-based salts all exhibit sufficient thermal stabilities for many applications, including battery electrolytes.

Transport properties

The ionic conductivity is an important parameter that allows us to assess the feasibility of ionic electrolytes for electrochemical applications. Fig. 3 shows the influence of different anions on the ionic conductivity of a range of [C₁moxa]⁺-based salts, as well as a comparison with [C₂moxa][FSI]. [C₁moxa][DCA] displays the highest ionic conductivity, of 1.4×10^{-2} S cm^{–1} at 30 °C, as this exists as a liquid over the temperature range studied. This value of ionic conductivity is close to that of [C₄mpyr][DCA] ($\sim 1.5 \times 10^{-2}$ S cm^{–1}),⁵¹ an IL that has established advantageous properties including supporting the reversible Zn^{2+/0} redox electrochemistry, for Zn redox flow cells.⁵²

For the OIPCs, [C₁moxa][BF₄] displays a distinct enhancement in ionic conductivity upon heating from 40 to 50 °C, across the II–I phase transition; this appears to be a common occurrence in other [BF₄][–]-based OIPCs such as [C₂epyr][BF₄].¹⁵ On the other hand, the ionic conductivity of [C₁moxa][TFSI] gradually increases by four orders of magnitude upon heating across the temperature range studied, as it transitions from a relatively low conductivity solid into a conductive liquid. [C₁moxa][PF₆] exhibits the lowest ionic conductivity, on the order of 10^{–9} S cm^{–1}, as observed in other [PF₆][–]-based OIPCs.^{15,20} The ionic conductivity of [C₂moxa][FSI] is higher than [C₁moxa][FSI] by one order of magnitude, demonstrating a benefit of the ethyl chain substituent. Further discussion of cation/anion dynamics in the OIPCs in relation to the ionic conductivity is given below.

Importantly, the ionic conductivity of these oxazolidinium-based salts is higher than the equivalent pyrrolidinium-based

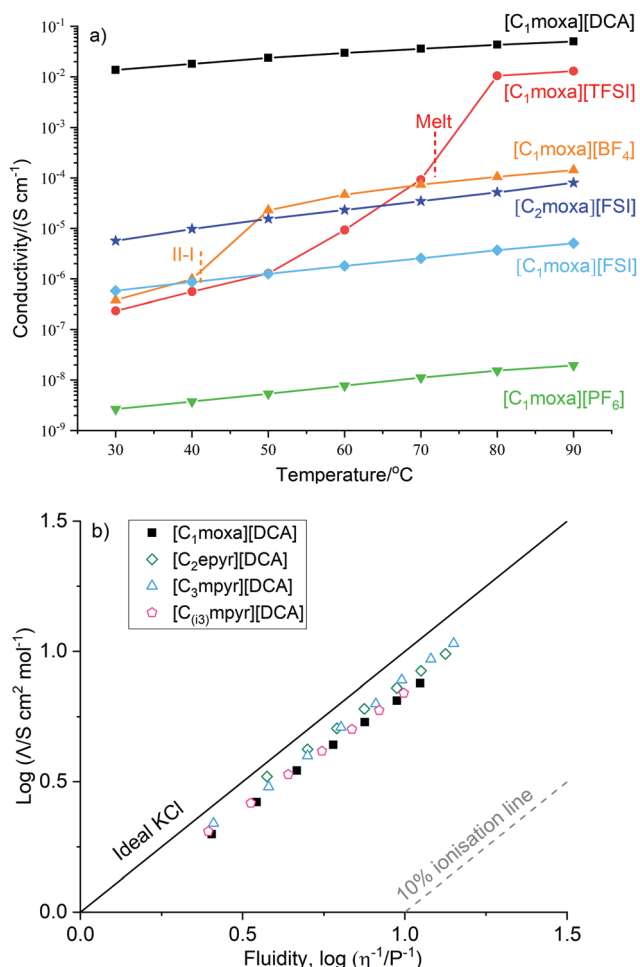


Fig. 3 (a) Plot of ionic conductivity of [C₁moxa]⁺-based salts and [C₂moxa][FSI], with the phase transition temperatures for [C₁moxa][BF₄] and [C₁moxa][TFSI] marked, and (b) Walden plot for [C₁moxa][DCA] from 30 °C to 90 °C. Comparison with ionicity values for [C₂mpyr][DCA],¹⁵ [C₃mpyr][DCA]⁶⁷ and [C₁₃mpyr][DCA]²¹ from the literature are shown.

salts (Table S1, ESI[†]). For example, at 30 °C and in phase I, the ionic conductivities of the OIPCs [C₁moxa][FSI] and [C₁mpyr][FSI] are 5.8×10^{-7} and 2.0×10^{-8} S cm⁻¹ respectively, and similarly the [C₂moxa][FSI] is around three times more conductive than [C₂mpyr][FSI] (6×10^{-6} S cm⁻¹ and 2×10^{-6} S cm⁻¹ respectively). In contrast, for oxazolidinium and pyrrolidinium-based room temperature ILs, the former displays slightly lower ionic conductivities and higher viscosities.³² For example at 25 °C, the previously reported ILs [C₄mpyr][TFSI] and [C₄moxa][TFSI] have ionic conductivities of 3×10^{-3} and 2×10^{-3} S cm⁻¹ respectively, at 25 °C.³² Therefore, there is a contrast in the conductivity trends observed depending if the salt is a solid or a liquid (OIPC or IL). This likely reflects the differences in the interactions and transport mechanisms in solids and liquids, and the impact of the ether functionality on these properties. Nevertheless, for the oxazolidinium-based OIPCs reported here, the presence of the ether group clearly presents an advantage for increasing the ionic conductivity.

The ionicity of [C₁moxa][DCA] was qualitatively assessed by examining the relationship between molar conductivity (Λ) and

fluidity ($1/\eta$) by using a Walden plot. This relationship compares the degree of dissociation of the cations and anions in the IL to an ideal reference electrolyte of aqueous KCl (0.01 mol L⁻¹), which is considered to be completely dissociated.^{53,54} Here, [C₁moxa][DCA] is plotted in comparison to other [DCA]-based ILs, shown in Fig. 3b. [C₁moxa][DCA] falls slightly below the ideal KCl line, suggesting that these ions tend to move almost independently from one another. At increased temperature, larger deviations from the ideal line are observed, reflecting an increase in interactions as found in many other ILs.^{55,56} Direct comparison to [C₁mpyr][DCA] is not possible due to its high melting point at 115 °C.⁵⁷ Nonetheless, the comparison to other pyrrolidinium [DCA]⁻-based ILs show that the salts containing cations with longer alkyl chains slightly exhibit more ion dissociation.

Solid-state NMR analysis

Solid-state NMR spectroscopy is a valuable technique used to assess the molecular level dynamics of the cations/anions in OIPCs. From the peaks in the static solid-state NMR spectra, the Full Width at Half Maximum (FWHM) values can provide information about the overall mobility of ions containing NMR active nuclei across a given temperature range; narrow linewidths indicate the presence of faster dynamics that more effectively average out anisotropic interactions.⁵⁸ As ¹H and ¹⁹F nuclei are present only in the cations and anions respectively, these species can be analysed separately. The static NMR spectra and temperature dependent FWHM data for the four more conductive OIPCs are presented in Fig. 4 and 5, respectively.

For the oxazolidinium-based salts studied, the largest increases in FWHM values are associated with phase transitions. For example, [C₁moxa][BF₄] exhibits two distinct increases in the ¹H and ¹⁹F FWHM values that correspond to the two solid-solid phase transitions near 41 °C and -7 °C. This indicates a decrease in dynamics of both the [C₁moxa]⁺ cations and [BF₄]⁻ anions, particularly at the II-I phase transition. The abrupt increase in ¹H FWHM values from 2 kHz to 16 kHz (Fig. 5a), and the increase in the ¹⁹F FWHM from 2 kHz to 12 kHz (Fig. 5b), indicates a substantial change in motion of both the cation and anion across the II-I phase transition. This correlates with the step increase in ionic conductivity observed upon heating across this phase transition (Fig. 3). In contrast, [C₁mpyr][BF₄] does not exhibit a sudden decrease in the ¹H FWHM values upon heating across the III-II phase transition at 65 °C,⁵⁹ rather a gradual decrease in NMR linewidths, which may reflect the more ordered, crystalline nature of that OIPC. While it would have been interesting to also determine the diffusion coefficients across these phase transitions, they were not measured due to the short *T*₂ values of these oxazolidinium-based salts.

[C₁moxa][FSI] displays a sharp increase in the ¹H FWHM values, from 8 kHz to 33 kHz, upon cooling through its II-I phase transition (Fig. 5a), indicating a decrease in cation dynamics. Note that as the NMR experiments were conducted by cooling from the highest temperature to the lowest, the phase transition would likely occur at more negative

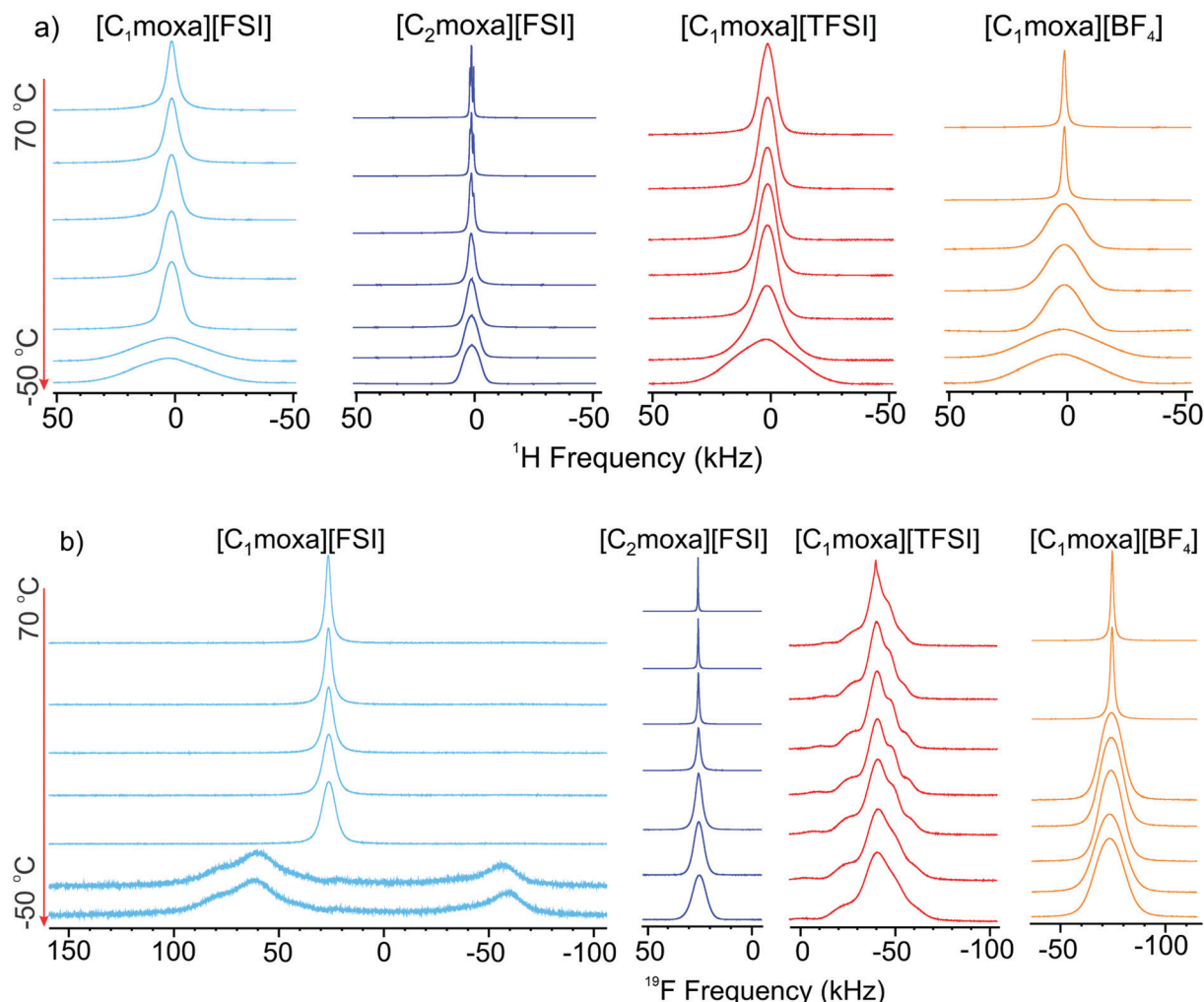


Fig. 4 (a) ^1H and (b) ^{19}F static NMR spectra of oxazolidinium-based OIPCs at temperatures of 70 °C, 50 °C, 30 °C, 10 °C, -10 °C, -30 °C and -50 °C.

temperatures than the transition temperature recorded during heating in the DSC.

Although this trend of decreased dynamics in phase II would also be expected for the $[\text{FSI}]^-$ anion in $[\text{C}_1\text{moxa}][\text{FSI}]$, Chemical Shift Anisotropy (CSA) effects are present in the spectra at -50 °C and -30 °C (Fig. 4b) such that the ^{19}F FWHM values cannot be determined quantitatively. Nevertheless, at a temperature above -10 °C, the ^{19}F linewidth is much smaller, decreasing from a broad asymmetric CSA line to a symmetric Gaussian line; this suggests an abrupt increase in anion dynamics in phase I. This is also consistent with the large ΔS of the II-I phase transition ($60 \text{ J K}^{-1} \text{ mol}^{-1}$), as $[\text{C}_1\text{moxa}][\text{FSI}]$ becomes substantially more disordered.

$[\text{C}_2\text{moxa}][\text{FSI}]$ displays a gradual increase in the ^1H and ^{19}F FWHM values as the temperature was decreased. Interestingly, the static ^1H NMR spectra of $[\text{C}_2\text{moxa}][\text{FSI}]$ (Fig. 4a) displays a broad peak and an additional narrow peak, suggesting that both relatively static and diffusive species are present, even at -50 °C. This line shape behaviour has been observed in other OIPCs, indicating the presence of multiple dynamic environments.^{40,60} Moreover, the proportion of this diffusive

component is larger at higher temperatures. For the $[\text{FSI}]^-$ anion the NMR linewidths are very narrow, particularly at temperatures above 30 °C where ^{19}F FWHM values are less than 1 kHz, indicating that the majority of the anions can exhibit translational diffusion.⁶¹ Comparison to $[\text{C}_1\text{moxa}][\text{FSI}]$ reveals that $[\text{C}_2\text{moxa}][\text{FSI}]$ exhibits more dynamics across the temperature range studied, evident by narrower NMR linewidths in both the cation and anion spectra and the presence of a narrow component in the ^1H NMR of the cation. These trends reflect an important advantage of utilising an ethyl substituent on the oxazolidinium cation; we hypothesise that the longer alkyl chain results in the formation of a less tightly-packed lattice, which is consistent with the lower T_m and higher conductivity observed. In contrast to the solid state NMR of $[\text{C}_2\text{mpyr}][\text{FSI}]$ measured by Nti *et al.*,⁶² the ^1H and ^{19}F spectra for $[\text{C}_2\text{moxa}][\text{FSI}]$ display a narrow component at -10 °C and -50 °C respectively, whereas for $[\text{C}_2\text{mpyr}][\text{FSI}]$ such narrow components are only present at temperatures above 50 °C. Hence, the NMR data indicates that both the cation and anion display higher mobility in $[\text{C}_2\text{moxa}][\text{FSI}]$ than in $[\text{C}_2\text{mpyr}][\text{FSI}]$, *i.e.* there are increased ion dynamics when ether functionality is

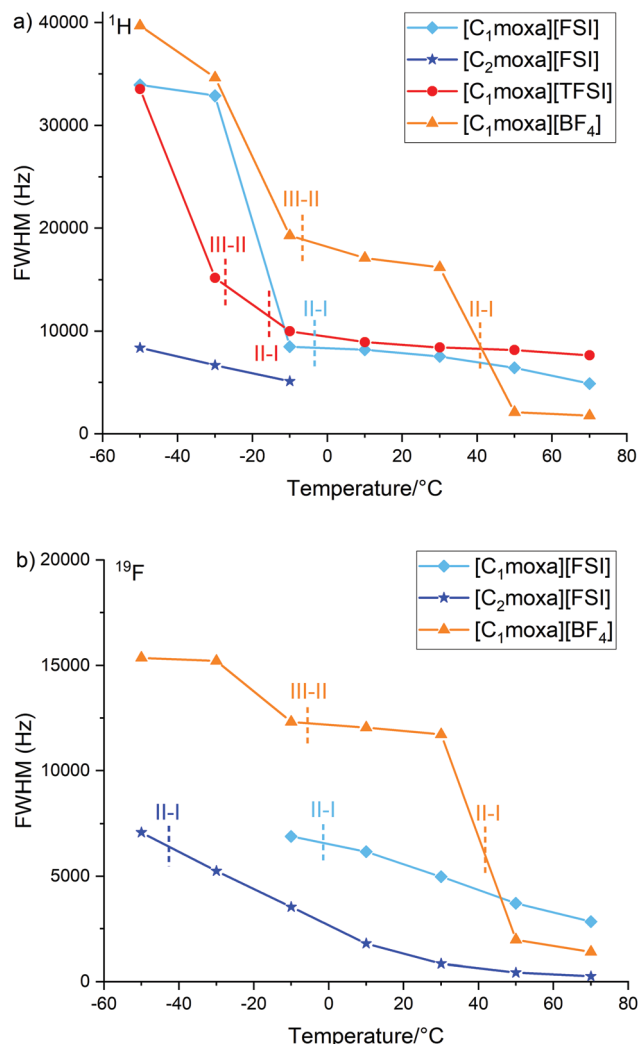


Fig. 5 NMR line widths for (a) ^1H and (b) ^{19}F , recorded at decreasing temperatures. The temperatures of the phase transitions of the different OIPCs are marked with a vertical dotted line.

present in the ring, which we tentatively attribute to a less tightly packed OIPC lattice and thus more free volume.

For $[\text{C}_1\text{moxa}][\text{TFSI}]$, a distinct increase in the ^1H FWHM values is seen upon cooling through the II-I (and III-II) phase transitions, as shown in Fig. 5a. On the other hand, the ^1H FWHM values are reasonably constant throughout phase I. As cation dynamics do not notably change over this temperature range (from -10 °C to 70 °C), the increase in conductivity that is

observed upon heating through phase I (Fig. 3) may be due more to increased dynamics in the anion than the cation. However, ^{19}F NMR linewidths of the $[\text{TFSI}]^-$ anion display asymmetric peaks that represent CSA patterns, likely due to the presence of multiple environments from the various dihedral conformers of $[\text{TFSI}]^-$,⁶³ and thus the FWHM cannot be quantified. Nevertheless, the peaks are clearly narrower at the higher temperatures than the lower (70 °C vs. -10 °C, Fig. 4b), consistent with more anion mobility.

Positron annihilation lifetime spectroscopy (PALS)

PALS analysis is used for the quantitative determination of free volume based within a material. The measured positronium lifetime (τ_3) represents the average size of pores/defects, whereas the intensity (I_3) describes the relative number of pores/defects in a given material. These defects provide valuable information as they are believed to be intrinsically involved in the mechanism of ionic conductivity in OIPCs.^{64,65} The Fractional Free Volume (FFV = average pore volume $\times$ intensity) takes into account both the size of pore defects and the number of pores, allowing for an overall comparison with other OIPCs. Table 2 summarises the PALS analysis for the oxazolidinium-based salts.

The FFV in $[\text{C}_2\text{moxa}][\text{FSI}]$ is greater than in $[\text{C}_1\text{moxa}][\text{FSI}]$ (977 vs. 885 respectively), where the lifetime and intensity are both larger, hence the ethyl chain on the cation appears to result in more, larger defects. This is consistent with the higher ionic conductivity of $[\text{C}_2\text{moxa}][\text{FSI}]$ at 30 °C. Whereas, the FFV in $[\text{C}_1\text{moxa}][\text{TFSI}]$ is the same, within uncertainty, as that of $[\text{C}_1\text{moxa}][\text{FSI}]$. This contrasts with previous studies on ammonium-based OIPCs containing $[\text{TFSI}]^-$ and $[\text{FSI}]^-$ anions,¹⁸ where much larger differences in free volume were observed between $[\text{N}_{1222}][\text{TFSI}]$ and $[\text{N}_{1222}][\text{FSI}]$ (146 and 93 respectively) when both are in phase I.¹⁸ Thus, the nature of the cation also plays a significant role in determining the FFV in the OIPC and the effect that the larger $[\text{TFSI}]^-$ anion has on the crystal packing and defects.

$[\text{C}_1\text{moxa}][\text{BF}_4]$ displays the lowest FFV – less than half of that in the imide-based salts. This is a reflection of the much smaller pore size (small lifetimes): it actually has a larger number of pores (*i.e.* greater intensities) compared to the imide-based salts. While the small pores may be attributed to the smaller and spherical nature of the anion, which could encourage closer packing, it is also important to note that $[\text{C}_1\text{moxa}][\text{BF}_4]$ is the only OIPC studied here that exists in phase

Table 2 PALS analysis data for oxazolidinium-based salts at 20 °C

	PALS analysis				
	Lifetimes	Intensities	Pore size	Pore	FFV (PALS)/a.u.
	τ_3/ns	$I_3/\%$	Diameter/nm	Volume/ $\text{\AA}^3$	
$[\text{C}_1\text{moxa}][\text{FSI}]$	2.061 ± 0.035	8.6 ± 0.2	0.582 ± 0.006	103 ± 1	885 ± 27
$[\text{C}_2\text{moxa}][\text{FSI}]$	2.138 ± 0.022	8.8 ± 0.1	0.595 ± 0.004	110 ± 1	977 ± 16
$[\text{C}_1\text{moxa}][\text{TFSI}]$	2.092 ± 0.022	8.6 ± 0.1	0.587 ± 0.004	106 ± 1	911 ± 14
$[\text{C}_1\text{moxa}][\text{BF}_4]$	1.397 ± 0.032	10.1 ± 0.4	0.444 ± 0.008	46 ± 1	463 ± 24

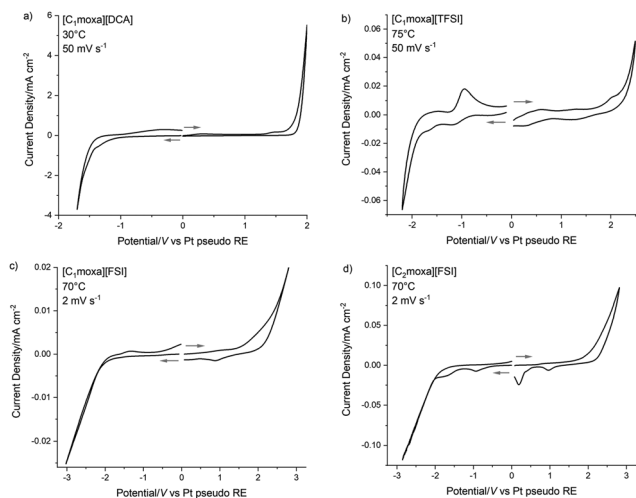


Fig. 6 Cyclic voltammetry of (a) $[C_1\text{moxa}][\text{DCA}]$ at 30 °C and scan rate of 50 mV s^{-1} , (b) $[C_1\text{moxa}][\text{TFSI}]$ at 75 °C and scan rate of 50 mV s^{-1} , (c) $[C_1\text{moxa}][\text{FSI}]$ and (d) $[C_2\text{moxa}][\text{FSI}]$ at 70 °C at a scan rate of 2 mV s^{-1} . All experiments were conducted with a Pt working electrode, Pt coil counter electrode and Pt wire pseudo reference electrode.

II at 30 °C. Thus, a substantial increase in total free volume may be expected upon heating across the II–I phase transition;⁶⁴ this would be consistent with the aforementioned increase in ion dynamics observed by NMR analysis, and the distinct jump in ionic conductivity after which the ionic conductivity of $[C_1\text{moxa}][\text{BF}_4]$ surpasses that of $[C_2\text{moxa}][\text{FSI}]$ at 50 °C. These variable temperature PALS studies will be the subject of future work.

In comparison to pyrrolidinium-based OIPCs, $[C_2\text{moxa}][\text{FSI}]$ (phase I) displays a similar FFV to $[C_2\text{mpyr}][\text{FSI}]$ (phase I)²⁰ (997 vs. 1008 respectively). Furthermore, $[C_1\text{moxa}][\text{TFSI}]$ (phase I) exhibits a free volume of 911, compared to $[C_1\text{mpyr}][\text{TFSI}]$ (phase II) of 835,⁶⁴ consistent with the higher ionic conductivity. Again, it is important to note that $[C_1\text{mpyr}][\text{TFSI}]$ exists in phase II at 30 °C. It is also observed that the oxazolidinium-based salts display higher lifetimes whilst exhibiting lower intensities, as seen with $[C_2\text{moxa}][\text{FSI}]$ and $[C_1\text{moxa}][\text{TFSI}]$ (2.14 ns, 8.8% and 2.09 ns, 8.6% respectively) compared to $[C_2\text{mpyr}][\text{FSI}]$ and $[C_1\text{mpyr}][\text{TFSI}]$ [1.92 ns, 11.2% and 1.72 ns, 10.8% respectively]. Hence, these oxazolidinium-based salts exhibit larger, yet a smaller number of, pores than equivalent pyrrolidinium-based salts, reflecting an influence of the ether group in the cation ring on crystal packing.

Electrochemical stability

The electrochemical stability of oxazolidinium-based salts was examined to assess their prospective use as electrolytes. The electrochemical potential windows were determined by CV on a Pt working electrode to measure the cathodic and anodic limit. Given that a Pt pseudo reference was used for initial investigations, it is the total electrochemical potential window that is the important parameter for considerations, rather than the specific cathodic/anodic potentials.⁶⁶ The CV of $[C_1\text{moxa}][\text{DCA}]$, $[C_1\text{moxa}][\text{TFSI}]$, $[C_1\text{moxa}][\text{FSI}]$ and $[C_2\text{moxa}][\text{FSI}]$ are shown in Fig. 6a–d.

$[C_1\text{moxa}][\text{DCA}]$ was found to exhibit an electrochemical potential window of ~ 3 V, which is the expected range of DCA-based ILs in the literature.^{67,68} The CV of $[C_1\text{moxa}][\text{TFSI}]$, conducted in liquid-state, displays an electrochemical potential window of ~ 4 V, where the oxidation peak near -1 V in the reverse scan is attributed to the reduction of decomposition products of the cation formed during the forward sweep. $[C_1\text{moxa}][\text{FSI}]$ and $[C_2\text{moxa}][\text{FSI}]$ both display relatively wide electrochemical potential windows of approximately 3.8 and 3.9 V respectively, showing only a small impact of alkyl chain length. Here, temperatures of 70 °C were used ensure that these OIPCs were soft enough to easily maintain contact between the OIPC electrolyte and the three-electrode setup. The peak near 0 V in the reverse scan of $[C_2\text{moxa}][\text{FSI}]$ is attributed to decomposition products formed during the forward sweep, which may be exacerbated by the high temperature and slow scan rate.

For comparison, the electrochemical stability of $[C_2\text{mpyr}][\text{FSI}]$ was measured using the same Pt three-electrode system, and exhibited an electrochemical potential window of 4.3 V (see Fig. S3, ESI†). Thus, $[C_2\text{moxa}][\text{FSI}]$ displays a narrower electrochemical potential window by ~ 0.4 V compared to $[C_2\text{mpyr}][\text{FSI}]$, attributed to the effect of the ether group in the ring. Such differences are similarly observed when ether functionality is present on the aliphatic alkyl chain of *N*-methoxyethyl-*N*-methylpyrrolidinium $[C_{201}\text{mpyr}][\text{FSI}]$, where authors attributed this to the electron withdrawing ether group that may facilitate the reduction of the cation.^{69,70} On the other hand, the electrochemical stability has been shown to improve through the addition of Li salts,^{71,72} e.g. Yamada *et al.* determined the electrochemical stability of a $[C_2\text{mpyr}][\text{FSI}]/\text{LiFSI}$ system to be ~ 5.4 V vs. Li/Li^+ .²² Thus, the initial assessment of oxazolidinium-based salts shows sufficient electrochemical stability to support further investigation into the benefits of these new OIPCs on Li electrochemistry, and extended studies into the development of new electrolytes for battery applications.

Conclusions

A range of oxazolidinium-based salts were synthesised and characterised to explore the effect of ether functionality in the ring, and to examine the influence of the anion on their physical and transport properties. The presence of an ether group in the ring lowered the melting point across all the salts studied, by more than 100 °C in some cases, compared to the pyrrolidinium analogues. $[C_1\text{moxa}][\text{DCA}]$ forms a room temperature IL, whereas $[C_1\text{moxa}][\text{TFSI}]$ is a low-melting OIPC. $[C_1\text{moxa}][\text{FSI}]$, $[C_2\text{moxa}][\text{FSI}]$, $[C_1\text{moxa}][\text{BF}_4]$ and $[C_1\text{moxa}][\text{PF}_6]$ are all higher-melting OIPCs with wide phase I ranges. $[C_1\text{moxa}][\text{FSI}]$ and $[C_2\text{moxa}][\text{FSI}]$ exhibited very low ΔS_f values (4 and 7 $\text{J K}^{-1} \text{mol}^{-1}$ respectively), reflecting the substantial disorder within these OIPCs.

Oxazolidinium-based salts were found to have higher ionic conductivities than the equivalent pyrrolidinium-based salts and $[C_2\text{moxa}][\text{FSI}]$ displayed the highest ionic conductivity, of $5.7 \times 10^{-6} \text{ S cm}^{-1}$ at 30 °C, out of the OIPCs studied. The solid-state NMR analysis revealed a temperature dependence of the

ion dynamics, with distinct increases in mobility associated with the phase transitions. [C₂moxa][FSI] displayed narrower ¹H and ¹⁹F NMR linewidths than [C₁moxa][FSI] across all temperatures, reflecting the influence of the ethyl side chain on the ion mobility of both the cation and anion. In addition, the presence of ether functionality in the ring resulted in an increase of pore/defect size. [C₂moxa][FSI] exhibited the largest free volume, consistent with its high ionic conductivity. Throughout the studies in this paper, [C₂moxa][FSI] displays the most promising electrolyte properties through its ionic conductivity, ion dynamics, and free volume. Future work will further explore the promise of this material as a lithium salt-containing electrolyte, where the presence of oxygen is predicted to have a beneficial effect on Li⁺ speciation.

Conflicts of interest

The authors have no conflicts of interest to declare.

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